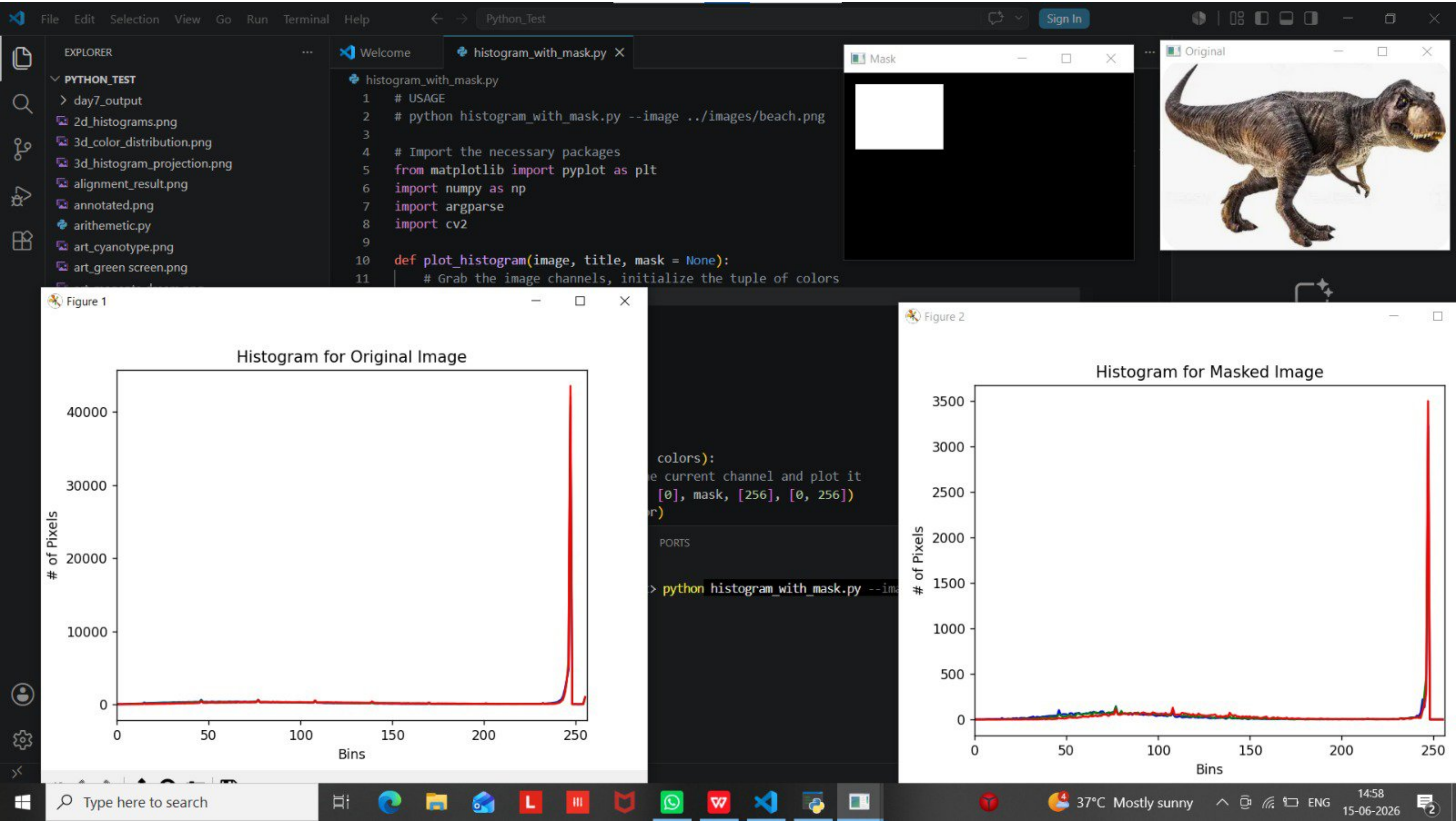
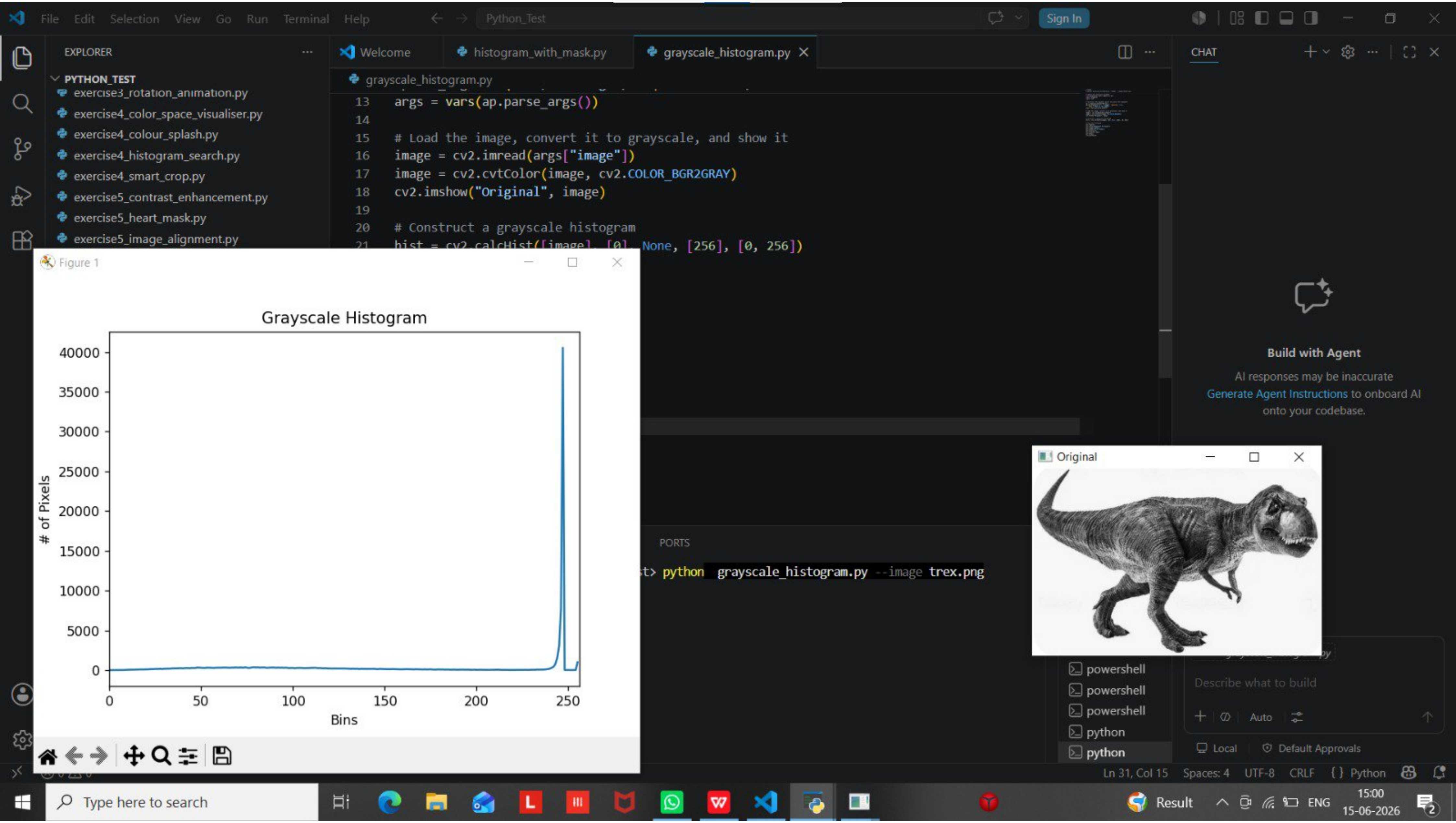
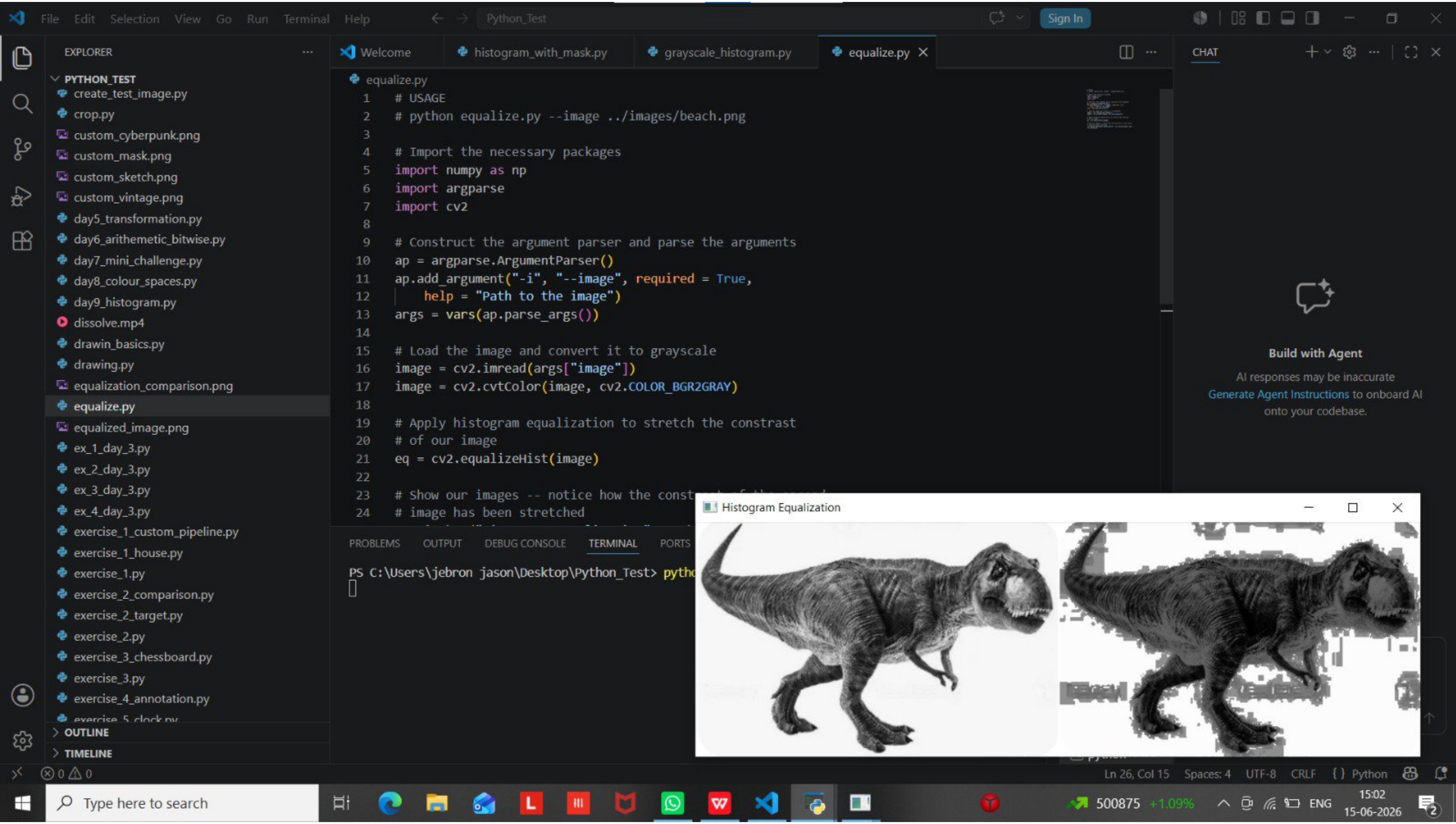
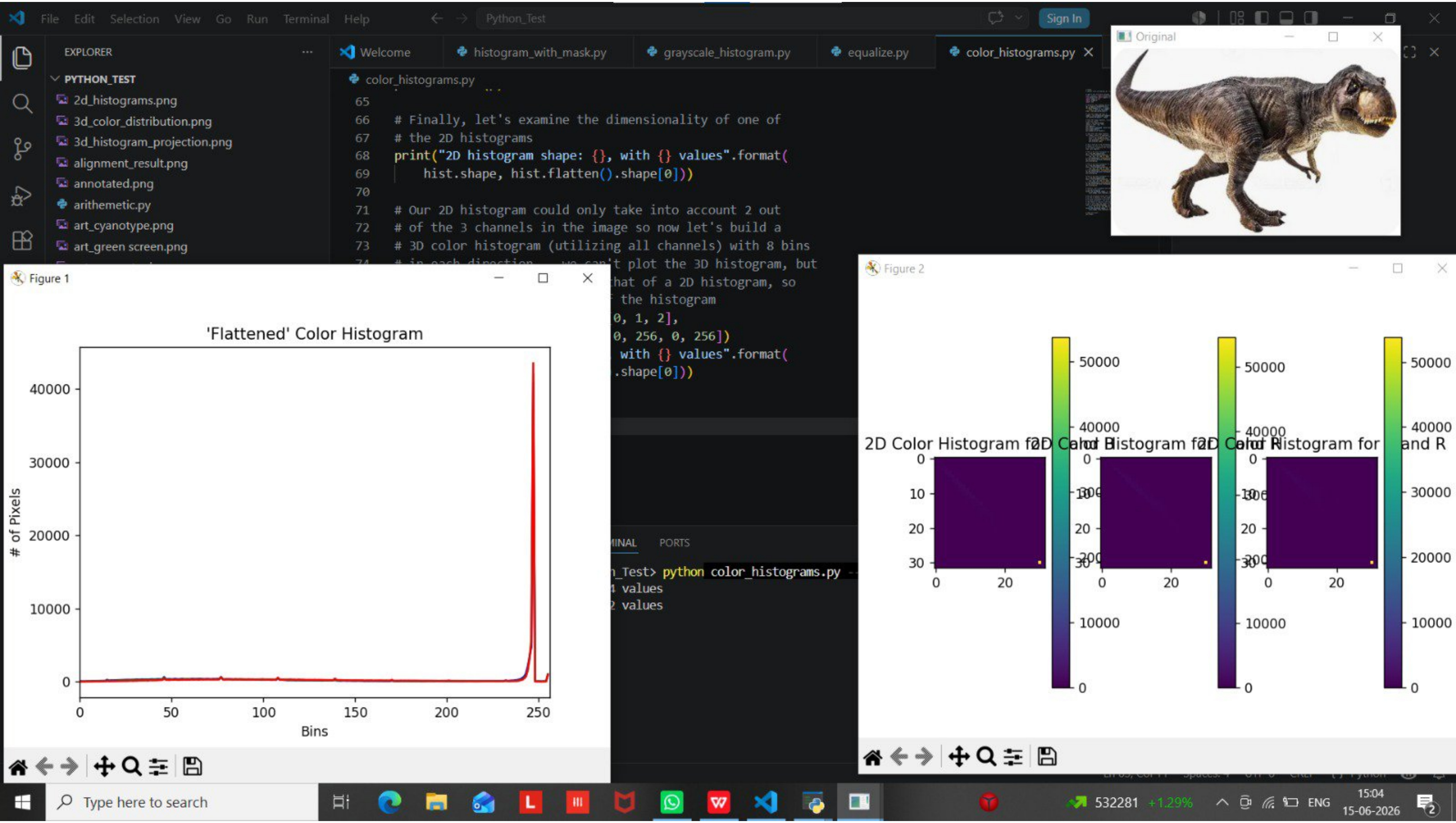

File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
day7_output
2d_histograms.png
3d_color_distribution.png
3d_histogram_projection.png
alignment_result.png
annotated.png
arithemetic.py
art_cyanotype.png
art_green screen.png
Welcome
histogram_with_mask.py
# USAGE
# python histogram_with_mask.py --image ../images/beach.png
# Import the necessary packages
from matplotlib import pyplot as plt
import numpy as np
import argparse
import cv2
def plot_histogram(image, title, mask = None):
# Grab the image channels, initialize the tuple of colors
colors):
e current channel and plot it
[0], mask, [256], [0, 256])
PORTS
python histogram_with_mask.py
Mask
Original
Figure 1
Histogram for Original Image
# of Pixels
Bins
Figure 2
Histogram for Masked Image
# of Pixels
Bins
Type here to search
37°C Mostly sunny
ENG
14:58
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise3_rotation_animation.py
exercise4_color_space_visualiser.py
exercise4_colour_splash.py
exercise4_histogram_search.py
exercise4_smart_crop.py
exercise5_contrast_enhancement.py
exercise5_heart_mask.py
exercise5_image_alignment.py
Welcome
histogram_with_mask.py
grayscale_histogram.py
args = vars(ap.parse_args())
# Load the image, convert it to grayscale, and show it
image = cv2.imread(args["image"])
image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
cv2.imshow("Original", image)
# Construct a grayscale histogram
hist = cv2.calcHist([image], [0], None, [256], [0, 256])
Figure 1
Grayscale Histogram
# of Pixels
Bins
0
5000
10000
15000
20000
25000
30000
35000
40000
0
50
100
150
200
250
PORTS
python grayscale_histogram.py --image trex.png
Original
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
powershell
powershell
powershell
python
python
Ln 31, Col 15
Spaces: 4
UTF-8
CRLF
{ } Python
Type here to search
Result
ENG
15:00
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
create_test_image.py
crop.py
custom_cyberpunk.png
custom_mask.png
custom_sketch.png
custom_vintage.png
day5_transformation.py
day6_arithemetic_bitwise.py
day7_mini_challenge.py
day8_colour_spaces.py
day9_histogram.py
dissolve.mp4
drawin_basics.py
drawing.py
equalization_comparison.png
equalize.py
equalized_image.png
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_custom_pipeline.py
exercise_1_house.py
exercise_1.py
exercise_2_comparison.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3.py
exercise_4_annotation.py
OUTLINE
TIMELINE
Welcome
histogram_with_mask.py
grayscale_histogram.py
equalize.py
# USAGE
# python equalize.py --image ../images/beach.png
# Import the necessary packages
import numpy as np
import argparse
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and convert it to grayscale
image = cv2.imread(args["image"])
image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply histogram equalization to stretch the constrast
# of our image
eq = cv2.equalizeHist(image)
# Show our images -- notice how the const
# image has been stretched
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> pytho
Histogram Equalization
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Ln 26, Col 15 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
500875 +1.09%
ENG
15:02
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
2d_histograms.png
3d_color_distribution.png
3d_histogram_projection.png
alignment_result.png
annotated.png
arithemetic.py
art_cyanotype.png
art_green screen.png
Welcome
histogram_with_mask.py
grayscale_histogram.py
equalize.py
color_histograms.py
Finally, let's examine the dimensionality of one of
the 2D histograms
print("2D histogram shape: {}, with {} values".format(
hist.shape, hist.flatten().shape[0]))
Our 2D histogram could only take into account 2 out
of the 3 channels in the image so now let's build a
3D color histogram (utilizing all channels) with 8 bins
Original
Figure 1
'Flattened' Color Histogram
# of Pixels
Bins
Figure 2
PORTS
python color_histograms.py
values
values
Type here to search
ENG
15:04
15-06-2026